wave function for a single particle, think of every location where the particle could possibly be found, were we to observe it. Then to each location assign an amplitude, a complex number with the property that the square of each number is the probability of finding the particle there. There is a constraint that the squares of all these numbers add up to precisely one, since the total probability that the particle is found somewhere must equal one. (Sometimes we speak of probabilities in terms of percentages, which are numerically 100 times the actual probability; a 20 percent chance is the same as a 0.2 probability.)

Notice we didn't mention "velocity" or "momentum" there. That's because we don't have to separately specify the momentum in quantum mechanics, as we did in classical mechanics. The probability of measuring any particular velocity is completely determined by the wave function for all the possible positions. Velocity is not a separate degree of freedom, independent of position. The basic reason why is that the wave function is, you know, a wave. Unlike for a classical particle, we don't have a single position and a single momentum, we have a function of all possible positions, and that function typically oscillates up and down. The rate of those oscillations determines what we're likely to see if we were to measure the velocity or momentum.

Consider a simple sine wave, oscillating up and down in a regular pattern throughout space. Plug such a wave function into the Schrödinger equation and ask how it will evolve. We find that a sine wave has a definite momentum, with shorter wavelengths corresponding to faster velocity. But a sine wave has no definite position; on the contrary, it's spread out everywhere. And a more typical shape, which is neither localized at one point nor spread out in a perfect sine wave of fixed wavelength, won't correspond to either a definite position or a definite momentum, but some mixture of each.